

Chapter 46: Postterm Pregnancy — Revision Notes

Williams Obstetrics 26e, pp. 2119–2140 of the PDF. High-yield numbers are in **bold**. The two tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 5 figures are included from the PDF.

Big picture: Perinatal risk is lowest at **39–40 weeks** and climbs after the due date. The dangers to the postterm fetus are **oligohydramnios** → **cord compression**, **thick meconium**, **growth restriction** and **macrosomia**. Accurate **first-trimester dating** prevents many "false" postterm diagnoses. **Induce after 42⁰/₇ wk**; at 41 wk either induce or start fetal surveillance.

1. Definitions

Term	Definition
Postterm (prolonged) pregnancy	>42 ⁰ / ₇ wk, i.e., ≥294 days from the first day of the LMP = 42 completed weeks
Late term	41 ⁰ / ₇ –41 ⁶ / ₇ wk (in the 42nd week, but 42 weeks not yet completed)
Postmature	Reserved for the uncommon clinical neonatal syndrome of pathologically prolonged pregnancy
"Postdates"	Avoided — the real problem is often an uncertain EDD

- Williams prefers **postterm** or **prolonged** pregnancy.

2. Estimated Gestational Age

- The definition assumes **ovulation 2 weeks after the LMP** → menstrual recall errors or delayed ovulation create false postterm diagnoses.
- **First-trimester sonography is the most accurate** way to establish or confirm GA.
- Reconcile LMP and first accurate ultrasound (Table 14-1) and **record the EDD**.

3. Incidence and Risk Factors

- US 2019: **0.3%** of 3.75 million births at **≥42 wk** — falling because of better dating and earlier intervention.
- **Risk factors:**
 - **Nulliparity** and **prepregnancy BMI ≥25** (the only significant factors in the Danish Birth Cohort)
 - **Long midpregnancy cervical length** (3rd/4th quartile) in nulliparas → **2×** likely to deliver after 42 wk
 - **Prior postterm birth** — familial tendency (mother and daughter); **maternal, not paternal, genes**; locus on **chromosome 2q13**
- Adverse outcomes rise with **advancing maternal age**.
- **Rare fetal-placental causes: anencephaly, fetal adrenal hypoplasia, X-linked placental sulfatase deficiency** (all ↓ estrogen precursors — Ch 5).

4. Perinatal Mortality and Morbidity

- **Stillbirth, neonatal death and infant morbidity all rise after the EDD.**
- Swedish data: perinatal mortality **nadir at 39–40 wk**, rising thereafter (same trend in the US).

- Main morbidities: **gestational hypertension, prolonged labor with CPD, birth injury, hypoxic-ischemic encephalopathy.**
- Slight ↑ **cerebral palsy** and **2-point lower IQ**; **no link with autism.**

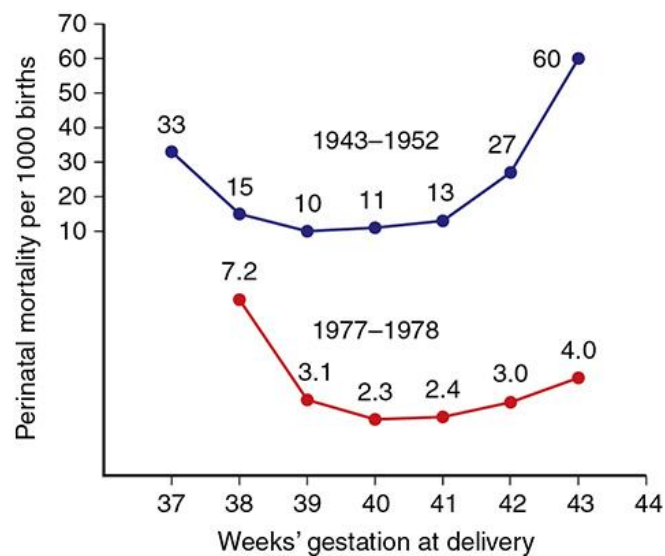


Figure 46-1. Swedish perinatal mortality is U-shaped with a nadir at 39–41 wk: 1943–52 rose from 13 (41 wk) to 60/1000 (43 wk); 1977–78 from 2.3 (40 wk) to 4.0/1000 (43 wk).

Table 46-1. Adverse Maternal and Perinatal Outcomes Associated with Postterm Pregnancy

Maternal	Perinatal
Fetal macrosomia	Stillbirth
Oligohydramnios	Postmaturity syndrome
Preeclampsia	NICU admission
Cesarean delivery — labor dystocia; fetal jeopardy	Meconium aspiration syndrome
Shoulder dystocia	Neonatal convulsions
Postpartum hemorrhage	Hypoxic-ischemic encephalopathy
Forceps delivery	Birth injuries
Perineal lacerations	Infection
	Childhood obesity

NICU = neonatal intensive care unit. From Amark, 2021; Kortekaas, 2020; Lindquist, 2021; MacDorman, 2009; Maoz, 2019; Middleton, 2018; Olesen, 2003.

Parkland (Alexander 2000a; 56,317 singletons ≥40 wk)

- Labor induced in **35%** of pregnancies reaching 42 wk.
- At **42 wk**: ↑ cesarean for dystocia or fetal distress; ↑ NICU admission; **neonatal seizures and deaths doubled.**

The denominator problem (Smith 2001)

- The population at risk in a given week = **all ongoing pregnancies**, not just births that week.
- Using this cumulative "perinatal index", **38 wk** had the lowest risk of perinatal death.
- In practice, death risk is balanced against immaturity → **no elective delivery before 39 wk** unless a comorbidity warrants it.

5. Pathophysiology

Postmaturity syndrome

- **Features:**
 - Wrinkled, patchy, **peeling skin**, most prominent on **palms and soles**
 - **Long, thin body** suggesting wasting
 - Advanced maturity: **open-eyed, unusually alert, "old and worried"** look
 - **Long nails**
- Most are **not** technically growth restricted (birthweight seldom <10th percentile), but severe FGR may be present.
- Complicates **10–20%** of pregnancies at 42 completed weeks.
- **Oligohydramnios** raises the likelihood: **88%** postmature if the maximal vertical pocket was **≤1 cm** at 42 wk (Trimmer 1990).



Figure 46-2. Postmaturity syndrome at 43 weeks: thick meconium on desquamating skin, long thin body, wrinkled hands.

Placental dysfunction

- Proposed: limited placental capacity with **dysfunctional syncytiotrophoblast** (Redman and Staff).
- Clifford (1954): skin changes from loss of protective **vernix caseosa**; "placental senescence" — but **no consistent histological or quantitative placental findings**.
- **Placental apoptosis** ↑ at 41–42 wk vs 36–39 wk; proapoptotic genes (e.g., **kisspeptin**) upregulated. Clinical significance unclear.
- **Cord erythropoietin** ↑ at **≥41 wk** (only stimulus is ↓ P_{O_2}) despite normal Apgar and acid–base → fetal oxygenation reduced in some.
- But fetuses **keep growing** (often large at birth) → placental function not severely compromised.
 - Growth velocity **peaks ~37 wk**, then slows; growth continues to at least **42 wk**.
 - **Umbilical blood flow does not increase** accordingly.

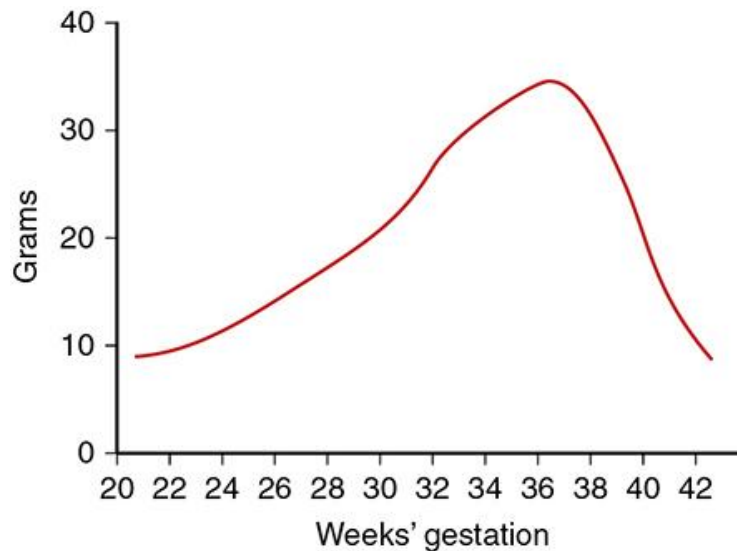


Figure 46-3. Mean daily fetal weight gain peaks at ~35 g/day around 37 weeks, then falls steeply (~10 g/day by 42 weeks).

Fetal distress and oligohydramnios

- Leveno 1984 (727 postterm pregnancies): antepartum and intrapartum jeopardy were due to **cord compression from oligohydramnios, not uteroplacental insufficiency** (no late decelerations).
- **Prolonged decelerations** preceded $\frac{3}{4}$ of emergency cesareans for nonreassuring FHR; **variable decelerations** were present in nearly all.
- **Saltatory baseline** (oscillations **>20 bpm**) was common — not ominous alone.
- Other correlates: oligohydramnios, **viscous meconium**, **nuchal cord**.
- Amnionic fluid volume normally **declines after 38 wk**. Meconium into scant fluid → thick → **meconium aspiration syndrome**; risk **5% at 42 wk**.

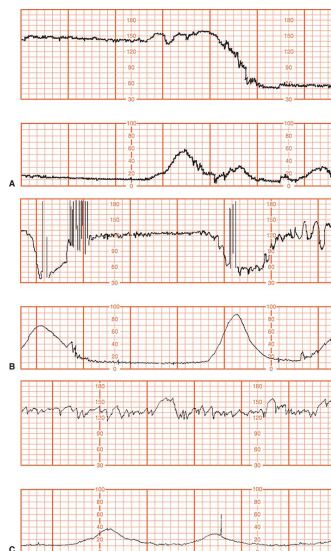


Figure 46-4. Postterm FHR patterns with oligohydramnios: (A) prolonged deceleration before emergency cesarean, (B) variable decelerations, (C) saltatory baseline >20 bpm — all point to cord compression.

Fetal-growth restriction

- Stillbirths are more common among growth-restricted fetuses delivered after 42 wk: **$\frac{1}{3}$ of postterm stillbirths were growth restricted** (Sweden).
- Parkland (355 neonates ≥ 42 wk and **<3rd percentile**): ↑ morbidity and mortality; this small group accounted for **$\frac{1}{4}$ of postterm stillbirths**.

6. Complications

Oligohydramnios

- Cause: ↓ **fetal urine production** (sequential bladder volumes) and ↓ **fetal renal blood flow** (Doppler), possibly from the lack of rising umbilical flow.
- Measured by **deepest vertical pocket (DVP)** or **AFI** (Ch 14); no exact definition.
 - The smaller the pocket, the greater the risk — but a **normal volume does not exclude** a bad outcome (oligohydramnios can develop quickly).
 - RCT (500 women): **AFI overestimates** abnormal outcomes compared with DVP.
- **Oligohydramnios by most definitions is clinically meaningful** → more intrapartum "fetal compromise".

Macrosomia

- Weight gain velocity peaks at **~37 wk** but most fetuses keep gaining; **95th percentile at 42 wk = 4475 g**.
- Brachial plexus injury was **not** related to postterm gestation (one study).
- **Induction for suspected macrosomia is not supported** (ACOG).
- Without diabetes, **vaginal delivery is not contraindicated up to EFW 5000 g** (Ch 27). Fetal weight estimates vary widely.

7. Antepartum Management

- **After 42 completed weeks → labor induction recommended.**
- For late term (41 wk), some intervention is indicated; induction vs expectant management with surveillance is debated.
- Older ACOG survey: **73%** routinely induced at 41 wk; most others did **twice-weekly testing** until 42 wk.

Induction factors

- **Unfavorable cervix: Bishop score <7** in **92%** at 42 wk (Harris 1983); **40%** "undilated" at 41 wk.
- **No cervical dilation → 2× cesarean for "dystocia"** (Parkland, 800 inductions).
- **Transvaginal cervical length ≤3 cm** predicts successful induction; **≤25 mm** predicts spontaneous labor or successful induction.
- **Prostaglandins:** PGE₂ gel was **not better than placebo** (MFMU 1994). With 2–4 cm dilation, almost **half** entered labor with PGE₂ alone.
- **Membrane sweeping:** slightly ↑ spontaneous labor and ↓ induction (Cochrane, 44 studies) but **no ↓ cesarean**; other trials no ↓ induction. Downsides: pain, bleeding, irregular contractions.
- **Station predicts cesarean** in nulliparas induced after 41 wk (Shin 2004):

Station before induction	Cesarean rate
–1	6%
–2	20%
–3	43%
–4	77%

Induction versus fetal testing

Study	Design	Result
Hannah 1992 (Canada)	3407 women ≥ 41 wk: induction vs testing	Induction ↓ cesarean 21% vs 24% ; only 2 stillbirths with testing
MFMU (Gardner 1996)	41 wk: induction (265) vs twice-weekly NST + AF volume (175)	No perinatal deaths; cesarean rate the same → both strategies valid
Alexander 2001	≥ 41 wk	Spontaneous labor before 42 ⁰ / ₇ wk vs induction after 42 ⁰ / ₇ wk: cesarean 14% vs 19% (failure to progress)
Denmark (Zizzo 2017)	Guideline changed 2011: induction at 41²/₇–41⁶/₇ wk + surveillance from 41 ⁰ / ₇ wk (was 42 ⁰ / ₇ wk, no surveillance)	Pregnancies past 42 wk 2.85% → 0.62% ; perinatal mortality 22 → 13 per 1000 ; cesarean rate unchanged
Bleicher 2017	Before–after	Induction at ≥ 42 wk: cesarean 15% vs 19.4%
SWEPIs (Wennerholm 2019)	2760 low-risk women: induction at 41 wk vs expectant + induction at 42 wk	Primary composite no different ; stopped early for ↑ perinatal mortality with expectant care (0 vs 6/1000); ↑ macrosomia with expectant care

Table 46-2. Selected Maternal and Perinatal Outcomes from SWEPIs Trial

Outcome	Induction (n = 1381)	Expectant (n = 1379)	p value
<i>Maternal</i>			
Gestational age (days)	289 ± 1.3	292 ± 2.7	—
Labor (hours)	7.1 ± 5.4	8.3 ± 5.9	<0.001
Cesarean delivery	10.4%	10.7%	0.79
Operative vaginal delivery	6.4%	6.6%	0.87
<i>Perinatal</i>			
Primary composite	2.4%	2.2%	0.9
Perinatal mortality	0	6/1000	0.03
Meconium aspiration	0.1%	0.2%	1.00
Macrosomia (>4500 g)	4.9%	8.3%	<0.001
NICU stay (days)	3.4 ± 3	4.6 ± 5.6	NS

g = grams; NICU = neonatal intensive care unit; NS = not significant; SWEPIs = SWEdish Post-term Induction Study. Note: the text says only perinatal mortality and macrosomia differed, but the table also shows a significantly shorter labor with induction ($p < 0.001$).

Management strategies

- Evidence is **insufficient to mandate** one strategy between 40 and 42 completed weeks → **induction or fetal surveillance at 41 wk** are both reasonable.
- **ACOG/SMFM: fetal surveillance once or twice weekly from 41⁰/₇ wk.**
- **After 42 completed weeks → induce.**
- With the higher stillbirth rate on expectant management at 41 wk, **routine induction at 41 wk will likely become preferred.**
- **Uncertain GA:** deliver at **41 wk** by the best clinical estimate. **No amniocentesis for fetal lung maturity.**

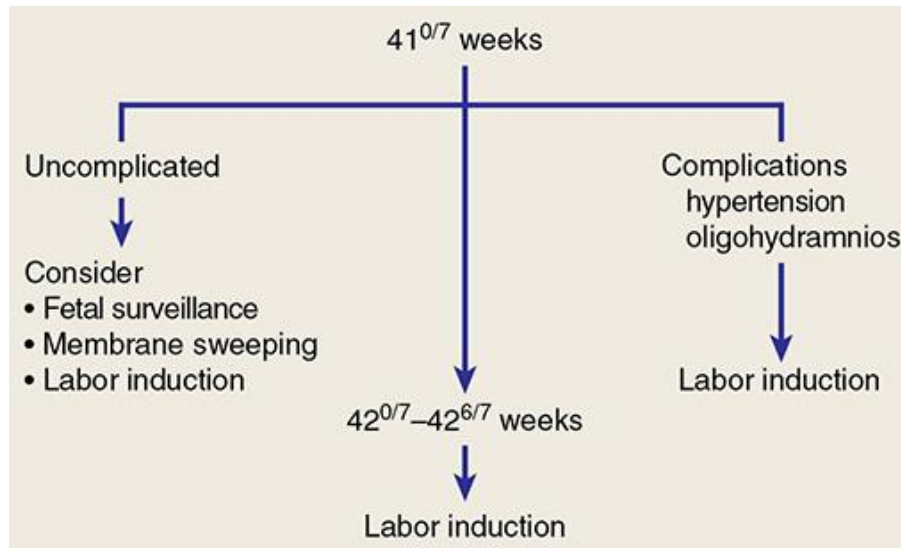


Figure 46-5. ACOG algorithm: at 41^{0/7} wk, uncomplicated → consider surveillance, membrane sweeping or induction; complications (hypertension, oligohydramnios) → induce; 42^{0/7}–42^{6/7} wk → induce.

Memory hook: "41 — choose; 42 — induce." Oligohydramnios or hypertension at any point after 41 wk → induce.

8. Intrapartum Management

- Labor is particularly dangerous for the postterm fetus → come to hospital **as soon as labor is suspected**; **continuous electronic FHR and contraction monitoring**.
- **Amniotomy dilemma:**
 - Against: further ↓ fluid → ↑ **cord compression**.
 - For: allows a **scalp electrode and IUPC** (more precise data) and reveals **thick meconium**.
- **Thick meconium** = little liquid (oligohydramnios); aspiration → severe pulmonary dysfunction and death.
- **Amnioinfusion does NOT prevent meconium aspiration syndrome**, but is reasonable for **repetitive variable decelerations**.
- **Do not suction the pharynx** at delivery of the head. **Intubate the depressed newborn** with meconium-stained fluid.
- **Nullipara in early labor with thick meconium** → vaginal delivery less likely → **strongly consider prompt cesarean**, especially with suspected **CPD** or **hypotonic/hypertonic dysfunctional labor**. Some avoid oxytocin.

Quick-Recall: Postterm Problems and Their Mechanism

Problem	Mechanism / Key fact	Clue / Action
False postterm	Dating error, delayed ovulation	First-trimester ultrasound dating
Oligohydramnios	↓ Fetal urine, ↓ renal blood flow	DVP (AFI overcalls); → induce
Nonreassuring FHR	Cord compression (not placental insufficiency)	Prolonged/variable decelerations, saltatory baseline
Meconium aspiration	Thick meconium in scant fluid (5% at 42 wk)	No pharyngeal suction; intubate depressed newborn; amnioinfusion does not prevent it
Postmaturity syndrome	10–20% at 42 wk; 88% if pocket ≤1 cm	Peeling skin, long thin body, alert "old" face, long nails
FGR	1/3 of postterm stillbirths; 1/4 at Parkland (<3rd %ile)	Highest-risk subgroup
Macrosomia	Growth continues; 95th %ile 4475 g at 42 wk	Induction for suspected macrosomia not supported; vaginal OK ≤5000 g without diabetes
Failed induction	Unfavorable cervix (Bishop <7 in 92%), high station	Station –4 → 77% cesarean

Key Numbers to Memorize

- Postterm **>42⁰/₇ wk / ≥294 days**; late term **41⁰/₇–41⁶/₇ wk**
- Incidence **0.3%** at ≥42 wk (US 2019)
- Perinatal mortality nadir **39–40 wk**; Smith's perinatal index lowest at **38 wk**
- Seizures and deaths **doubled at 42 wk** (Parkland)
- Postmaturity **10–20%** at 42 wk; **88%** if max vertical pocket **≤1 cm**
- Growth velocity peaks **~37 wk**; 95th percentile at 42 wk **4475 g**; vaginal delivery OK to **5000 g** (no diabetes)
- Meconium aspiration syndrome **5%** at 42 wk
- Emergency cesareans preceded by prolonged deceleration: **3/4**; saltatory baseline **>20 bpm**
- FGR = **1/3** of postterm stillbirths
- Bishop **<7** in **92%** at 42 wk; cervical length **≤3 cm / ≤25 mm** favors success
- Station –1/–2/–3/–4 → cesarean **6/20/43/77%**
- Hannah: cesarean **21 vs 24%**; Denmark perinatal mortality **22 → 13/1000**; SWEPIIS perinatal mortality **0 vs 6/1000**
- Surveillance **1–2x/week from 41⁰/₇ wk**; induce after **42 wk**; uncertain dates → deliver at **41 wk**